



Dungeon Escape

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Difficulty Level : Easy

Dungeon Escape

You awake in a damp, torchlit dungeon with no memory of how you got there. The only way out is through a maze of corridors, but some doors are locked. Rumour has it that a single rusty key lies somewhere in the dungeon, capable of opening every locked door. Time is of the essence — every step counts.

Task: Find the minimum number of steps required to reach the exit, or determine that escape is impossible.

Input

The first line contains two integers N and M ($1 \leq N, M \leq 500$) — the number of rows and columns in the dungeon grid.

The next N lines each contain a string of length M consisting of the following characters:

- `.` — an empty, passable cell
- `#` — a wall (impassable)
- `K` — a locked door; you may only pass through it if you are carrying the key
- `k` — the key; stepping on this cell automatically picks it up, permanently unlocking all locked doors
- `S` — your starting position (exactly one exists)
- `E` — the exit (exactly one exists)

You can move up, down, left, or right by one cell per step.

It is guaranteed that there is **at most one** key in the dungeon.

Output

Print a single integer — the minimum number of steps to reach `E` from `S`. If it is impossible, print `-1`.

Constraints

- $1 \leq N, M \leq 500$
- At most one `k` exists in the grid.
- Exactly one `S` and one `E` are present.
- Time limit: **1 second**
- Memory limit: **256 MB**

Sample Input 1

```
5 5
S...K
###.#
k...E
###.#
.....
```

Sample Output 1

6

Explanation

The grid contains a key and a locked door, but the shortest path from S to E does not require using them. The player can move right three times, down twice, and right once to reach the exit in 6 steps. The locked door K and key k are not used in this optimal route.

Sample Input 2

```
3 5
S...K
#k##E
...#.
```

Sample Output 2

7

Explanation

The grid contains a key and a locked door. The shortest path from S to E requires the player to first collect the key, then pass through the locked door. The player moves right to $(0, 1)$, down to $(1, 1)$ to pick up the key, back up to $(0, 1)$, right three times through $(0, 2)$, $(0, 3)$, and the now-unlocked door K at $(0, 4)$, and finally down to $(1, 4)$ to reach the exit in 7 steps. The locked door K blocks the direct top-row route, making the key mandatory for an optimal solution.